

Install low■drag helical rib turbulators (smooth■walled, optimized pitch and height) inside the tube, combined with a passive pulsating flow pattern (alternating flow direction every few minutes) to increase shear without adding significant pressure loss.

Executive Technology Brief

Domain: fluidics_hydraulic

Technology Maturity: ENGINEERING_DEFINITION

Transfer Posture: Sponsored Validation

Problem: Documented problem 'fouling' concerning passive countercurrent heat exchanger. User-submitted engineering problem (session input): "Independent audit case HX-2026-0910-B: Investigate whether a passive countercurrent heat exchanger can recover at least 60 percent of sensible heat from 40 C industrial wastewater at 2 L/min into 15 C clean inlet water while maintaining less than 10 kPa pressure drop, with fouling-prone suspended so". ROUTING: doe_osti: routing state SUSPENDED_RELEVANCE — measured relevance instability (R375: same fusion papers relevant 3 -> 0; pooled relevant-rate 0.175 pre-fix; DETERMINISM_GAP flagged in the R394 production audit) — records poison synthesis-feeding retrieval more than they feed it until relevance is semantically adjudicated; reinstatement: the Phase P1 semantic reranker lands (semantic relevance adjudication between query and record, replacing lexical overlap); flip to ACTIVE + a measured relevance re-run + worklog entry. [fra_rail_accidents] retrieval status GRAMMAR_MISMATCH (query does not match source grammar: YYYY-MM-DD date floor (\$where date >= X) — engine routing error, not a provider failure (not absence, Art. XXV)) — provider outcome recorded, not treated as absence (Art. XXI.3). [doe_osti] retrieval status NOT_QUERIED_SUSPENDED_RELEVANCE (engine routing decision (R399 W3): SUSPENDED_RELEVANCE — not a provider failure and not absence) — provider outcome recorded, not treated as absence (Art. XXI.3). Problem-existence basis: NO SOURCE RETURNED RECORDS — problem existence UNVERIFIED by live evidence (Art. XX); treat as hypothesis.

Mechanism: Vortex■induced shear from helical turbulator ribs creates secondary flows that continuously dislodge particles, reducing fouling while enhancing convective heat transfer, allowing higher heat recovery at low pressure drop.

Architecture: PROPOSED architecture (no physical system exists): flow path / lumen architecture; pressure regulation element; sensing / feedback element (if closed-loop); termination interfaces (patient / reservoir) — realizing: Install low■drag helical rib turbulators (smooth■walled, optimized pitch and height) inside the tube, combined with a passive pulsating flow pattern (alternating flow direction every few minutes) to increase shear without adding significant pressure loss.

Governing Model: Fluid mechanics / hydraulics governing relations proposed for this invention; SYMBOLIC ONLY until inputs are sourced

Key Equations:

FLUID-001: Hagen-Poiseuille (laminar pipe flow). Expression: $Q = (\pi * r^4 * \Delta P) / (8 * \mu * L)$.

Applicability to this invention: APPLICABLE; holds under: Newtonian fluid, laminar regime, rigid circular lumen, fully developed flow. Assumptions: CSF approximated as Newtonian (valid at shunt flow rates); no wall compliance effects. Engaged variables: ΔP .

FLUID-002: Reynolds number. Expression: $Re = (\rho * v * D) / \mu$. Applicability to this invention:

CONDITIONAL; holds under: circular section; single-phase flow. Assumptions: transition criterion $Re \sim 2300$ is a MODEL_DERIVED engineering rule of thumb. Engaged variables: NOT ESTABLISHED.

FLUID-003: Orifice flow (sharp-edged restriction). Expression: $Q = C_d * A * \sqrt{2 * dP / \rho}$. Applicability to this invention: APPLICABLE; holds under: turbulent orifice flow; C_d treated as MODEL_DERIVED until measured. Assumptions: $C_d \sim 0.6-0.65$ is an engineering prior, not a measurement. Engaged variables: dP .

Kill Condition: killer experiment fails: UNKNOWN (pre-registered pass/fail required before any transfer)

Current State: ENGINEERING_DEFINITION. No physical prototype. No clinical validation.

Disclosure: This dossier distinguishes established evidence from proposals and unresolved questions.